Correlated Beliefs and Lifecycle Behavior

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Whether beliefs offset or amplify each other
depends on their **joint distribution**!

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1. Survey economic beliefs across multiple domains in a representative US sample
2. Characterize the joint distribution & identify clusters of individuals w/ similar beliefs
3. Quantify implications for behavior and welfare using a lifecycle model:
   - **Positive:** correlations attenuate passthrough from beliefs to actions
   - **Normative:** correlations lower welfare cost of disagreement
   - **Normative:** correcting single beliefs can harm welfare belief patterns

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4. **Ongoing:** preparing survey w/ CB of Denmark to link w/ admin data

Literature: Subjective Expectations and Household Behavior

1. **Households' subjective expectations** (Manski 2004)

- Substantial heterogeneity and violations of full information rational expectations
  - E.g., Life expectancy (Hurd and McGarry 1995), employment (Mueller, Spinnewijn, and Topa 2021; Caplin et al. 2023), stock returns (Giglio et al. 2021), Inflation (DAcunto et al. 2024), Housing (Armona, Fuster, and Zafar 2019; Liu and Palmer 2023), etc.
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2. **Correlation between behavioral biases**

- Small literature studies the correlation between behavioral biases (Stango and Zinman 2023; Chapman et al. 2023)

Contribution: focus on expectations ($\neq$ preferences) & integrate with a lifecycle model

Outline

Consumption-Saving Decisions

- **Simple lifecycle model:** agent retires in period A_r and dies in period A

$$\max \sum_{t=0}^{A-1} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}$$

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- **Agents have different beliefs:** but is certain to be correct

Future income	Life expectancy	Risk-free rate	Inflation rate
$\widetilde{y} = (1+\epsilon_{\mathbf{y}})\, y$	$\widetilde{A} = (1+\epsilon_{\mathbf{A}})\, A$	$\widetilde{r} = (1+\epsilon_{\mathbf{r}})\, r$	$\widetilde{i} = (1+\epsilon_{\mathbf{i}})\, i$

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- **Impact on consumption:** for small ϵ, the consumption bias is:

$$\frac{\widetilde{c}_0 - c_0}{c_0} \approx \epsilon_{\mathbf{y}} - \phi_A A \epsilon_{\mathbf{A}} - (\phi_{PI}^r + \phi_r (EIS - 1))(r\epsilon_{\mathbf{r}} - i\epsilon_{\mathbf{i}})$$

where $\phi_A > 0$ captures the sensitivity to life-expectancy, $\phi_{PI}^r > 0$ the sensitivity of permanent income to the interest rate, and $\phi_r (EIS - 1)$ substitution and wealth effect

Framework Takeaways

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  Details

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Main takeaways:

- Beliefs may not be identified from actions alone: different beliefs can produce the same behavior (Manski 2004)
  $\rightarrow$ Motivates directly eliciting expectations!
- Key expectations for lifecycle saving/investing decisions:
  $\rightarrow$ Asset returns, inflation, labor market outcomes, life expectancy

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2. Home prices
3. Mortgage rates
4. T-bill rates
5. Inflation
6. Life expectancy
7. Wage growth
8. Job loss
9. Job finding
10. Covariances

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- Additional questions: demographics, employment, income, financial literacy (big 4), psychological outlook (Scheier and Carver 1985)

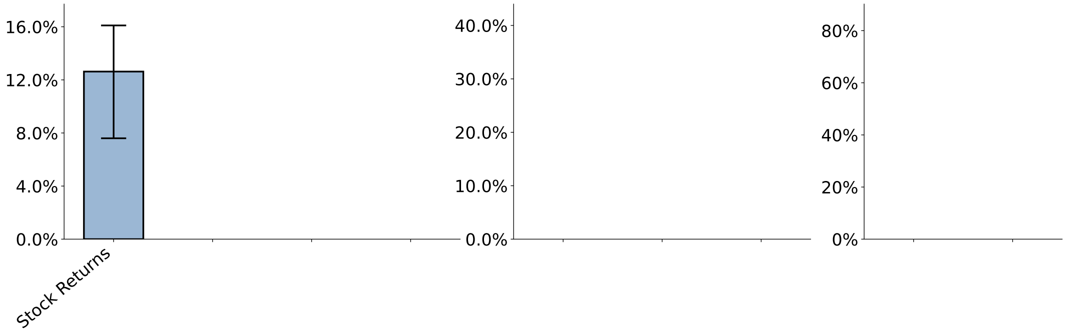
Survey Design

Imagine as a long term investment you invest in the **S&P 500 stock market index fund** (a fund that follows the performance of 500 of the largest publicly traded companies in the U.S). If you held that fund for the next **20 years**, what **average annual return** would you expect?

-50% -25% 0% 25% 50%

Avg. annual return

Average Expectations



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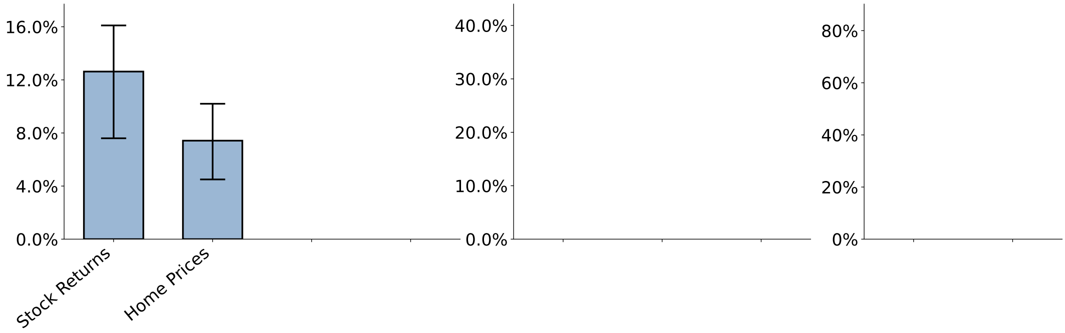

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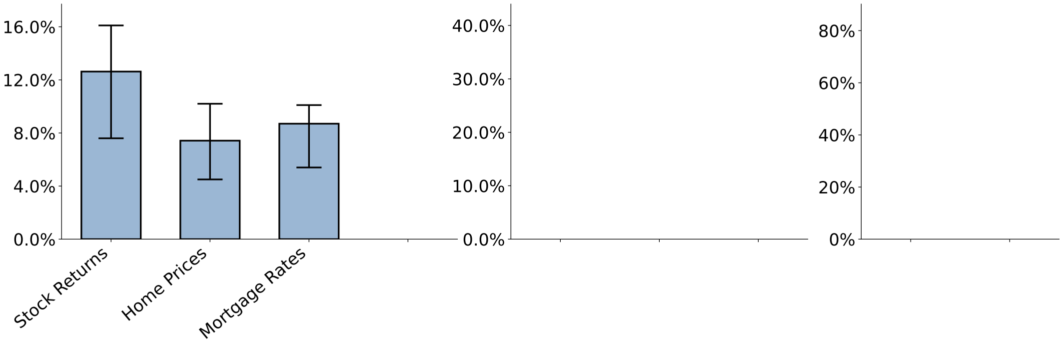

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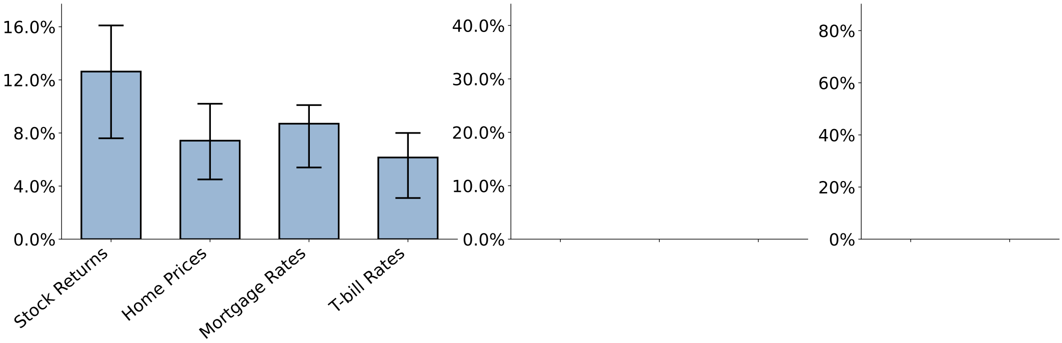

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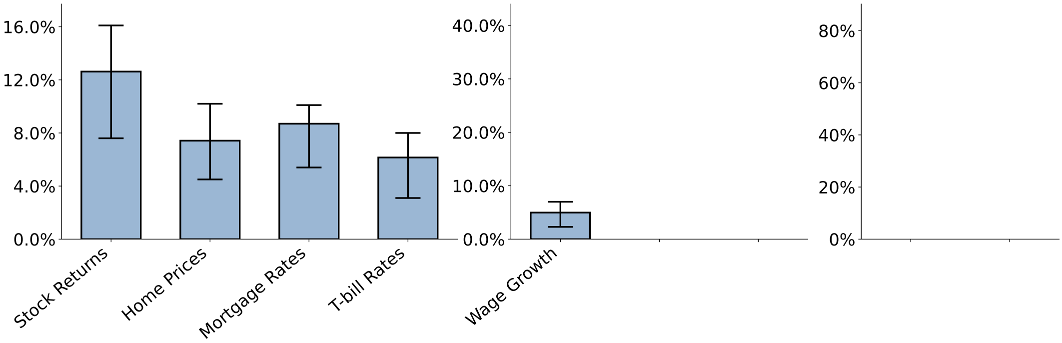

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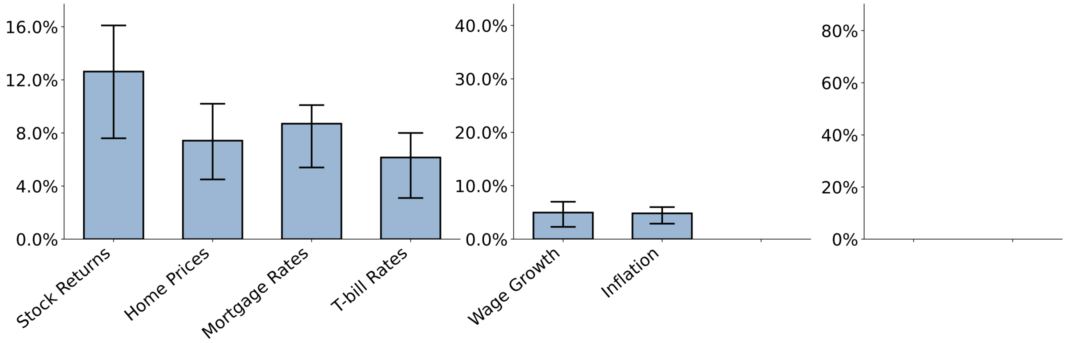

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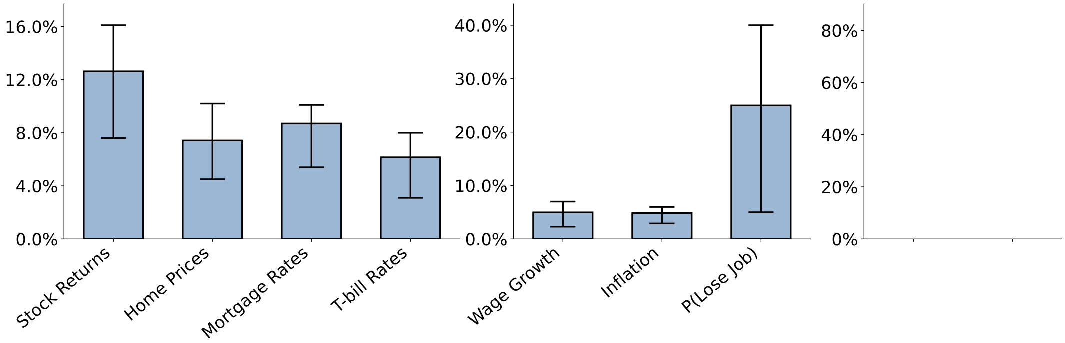

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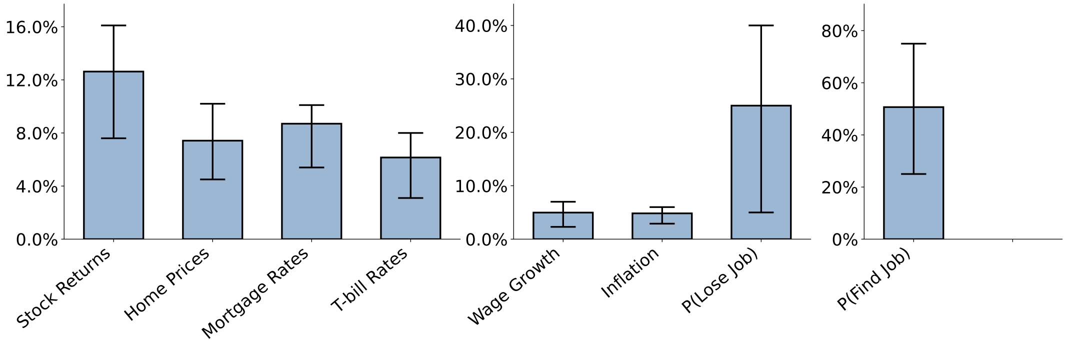

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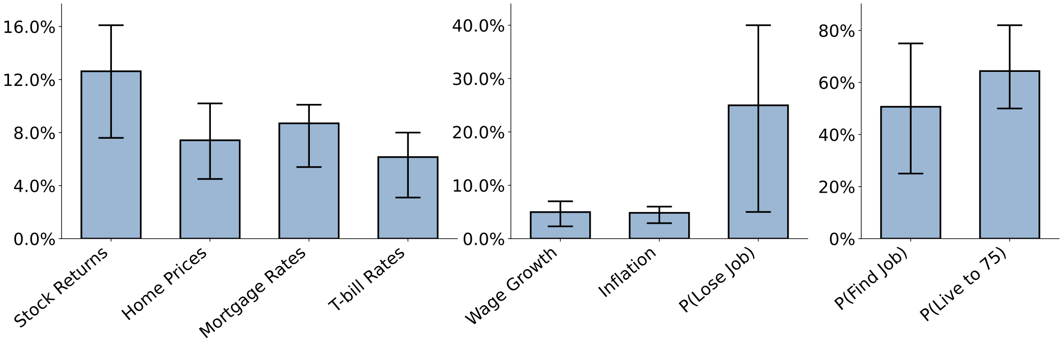

Age Income Employment Status Gender Race Financial Literacy

Measurement Error: Strategy

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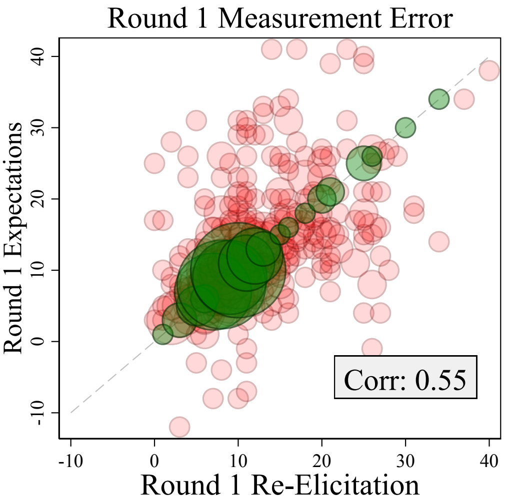
Classical error model: $x_{ir} = \mu_{ir} + \varepsilon_{ir}$

$$\underbrace{\mathrm{Var}(x_{ir})}_{\text{observed}} = \underbrace{\mathrm{Var}(\mu_{ir})}_{\text{true belief}} + \underbrace{\mathrm{Var}(\varepsilon_{ir})}_{\text{noise}}$$

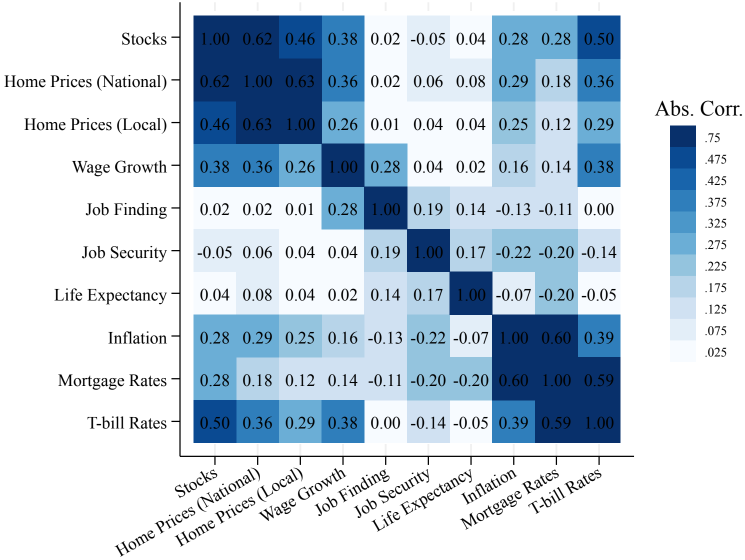
$$\widehat{\mathrm{Var}}(\varepsilon_{ir}) = \tfrac{1}{2}\mathrm{Var}(x_{ir} - x'_{ir})$$

Measurement Error: Result



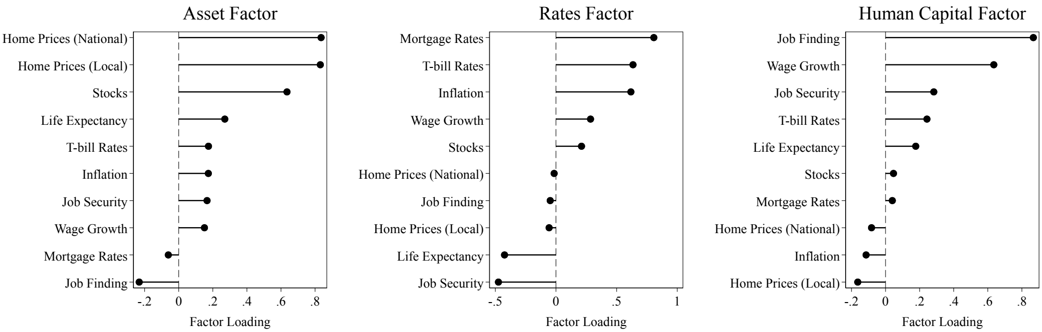
Confidence FinLit

Measurement Error Correction of Correlations



Outline

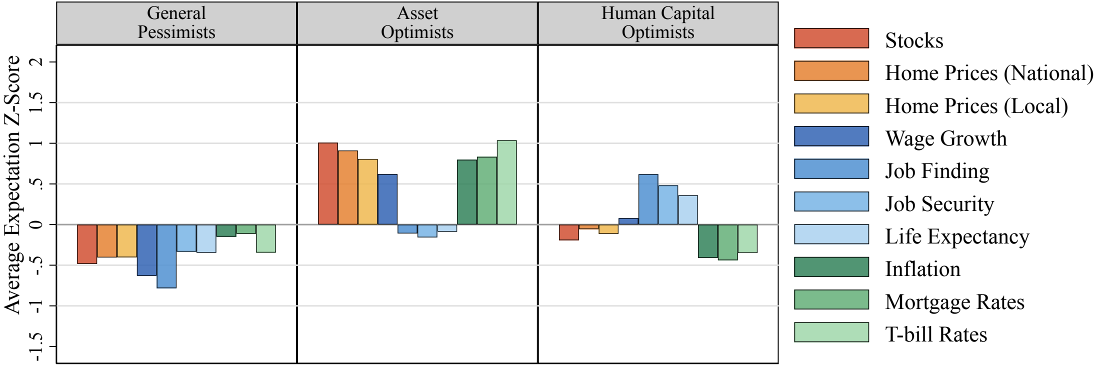
Fact 1. Beliefs co-move along three common factors



Common factor analysis after Stango and Zinman 2023; 3 factors $\Rightarrow$ 53% of common variance

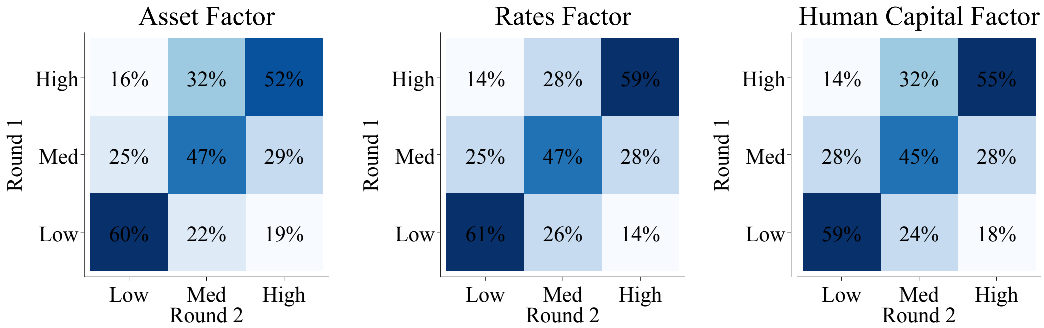
Asset factor (stocks + housing) | **Rates factor** (T-bill + mortgage + inflation) | **Human-capital factor** (wages + jobs + life expectancy)

Fact 2. Three belief types align with psychological optimism



General Pessimists	Asset Optimists	Human Capital Optimists
Age: 48.7	Age: 48.0	Age: 47.4
Income: $51,186	Income: $60,749	Income: $70,375
Fin. Lit.: 0.39	Fin. Lit.: 0.23	Fin. Lit.: 0.49
Optimism: 41.0%	Optimism: 54.9%	Optimism: 56.4%
Share: 37.3%	Share: 20.6%	Share: 42.2%

Fact 3. Beliefs are highly persistent within individuals



Tertile-to-tertile transitions across 6 months; persistence on the diagonal

- Diagonal share ≈ 50–60% across all three factors
- Transitions are *monotone* – long jumps are rare
- Once measurement error is removed: cross-round correlation **0.34** → **0.58**

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Lifecycle Model

Standard lifecycle framework: CRRA over consumption & housing services

- **Economic environment:**
  1. **Asset markets:** Housing (30y fixed mortgages), stocks, and one-period bond
  2. **Labor market:** Stochastic earnings (AR(1)) w/ unemployment and job-job transitions
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- **Agents decide each period:**
  1. **Consumption/saving:** how much to consume today vs save for the future?
  2. **Portfolio allocation:** What share of savings to hold in stocks vs one-period bond?
  3. **Housing:** Rent (flexible quantity) or own (subject to transaction cost)?

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4. **Group perceived covariances:** $\left\{\tilde{\rho}_j^{s,w}, \tilde{\rho}_j^{s,h}, \tilde{\rho}_j^{s,r}\right\}$

Model: investor's problem

Value function solved separately for each group j, taking $belief_j$ as fixed:

$$V_j(\boldsymbol{s}_{it}) = \max_{C_{it}, W_{it+1}, \alpha_{it}, house_{t+1}, H_{it+1}} U(C_{it}, H_{it}) + \beta\, \tilde{m}_{t,j}\, \mathbb{E}_j[V_j(\boldsymbol{s}_{it+1}) \,|\, \boldsymbol{s}_{it}, belief_j]$$

8 state variables $\boldsymbol{s}_{it}$:

- a_t: age (discrete)
- W_{it}: liquid wealth
- η_{it}: persistent earnings state
- emp_t: employment $\in \{E, U, Ret\}$
- $house_t$: tenure $\in \{O, R\}$
- H_{it}: housing quantity
- M_{it}: mortgage principal
- j_t: belief group (discrete)

Bringing the model to the data

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- **Simulation**:
  - agents face common realizations but use group-specific policy functions

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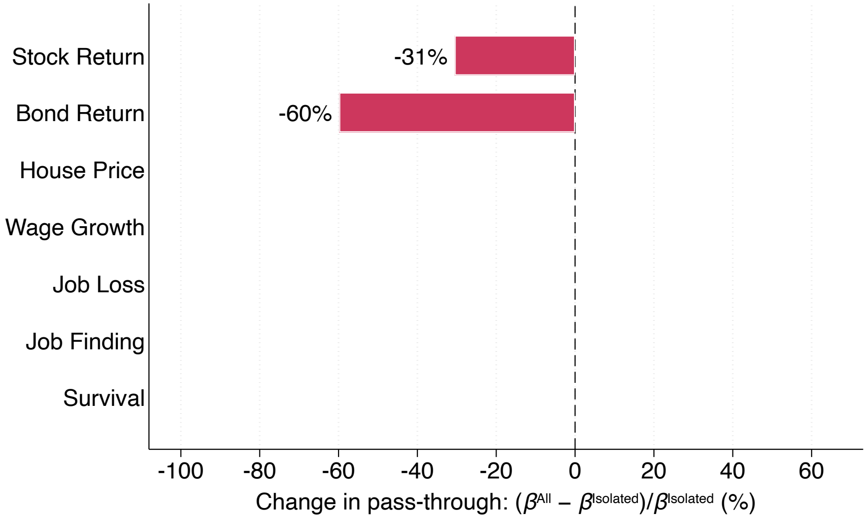
Compare regressions of behavior on beliefs across the two calibrations.

$\Rightarrow$ The difference measures the **omitted-variable bias** from studying expectations *one domain at a time*.

$$\underbrace{\tilde{\beta}_1}_{\text{isolated estimate}} = \underbrace{\beta_1}_{\text{true effect}} + \underbrace{\beta_2 \cdot \mathrm{Corr}(x_1, x_2) \cdot \frac{\sigma_{x_2}}{\sigma_{x_1}}}_{\text{omitted-variable bias}}$$

Finding 1: Correlations Attenuate Pass-through to Actions



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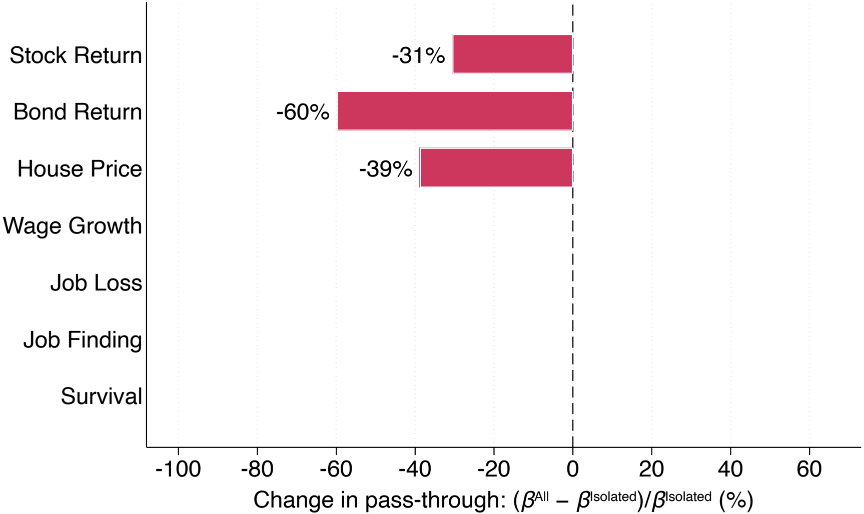

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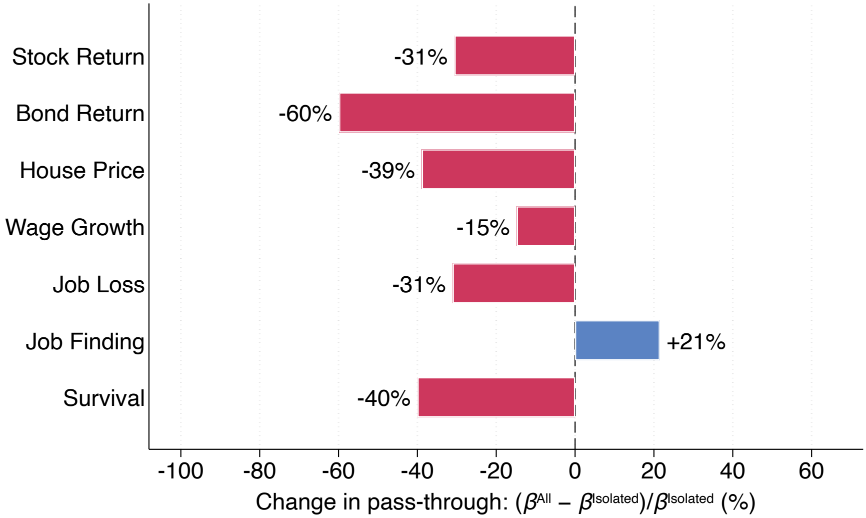
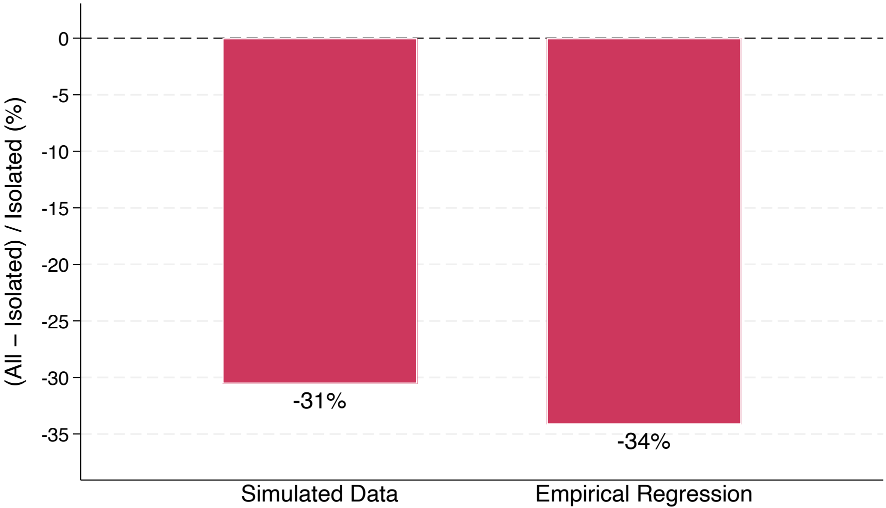


Finding 1: Correlations Attenuate Pass-through to Actions



Two Counterfactual Welfare Calculations

Compute lifetime consumption-equivalent welfare loss under:

1. **Independent beliefs.**
   All beliefs vary according to empirical dispersion, but no cross-domain correlations.
2. **Full joint distribution.**
   Same mean and dispersion, but matching observed correlations.

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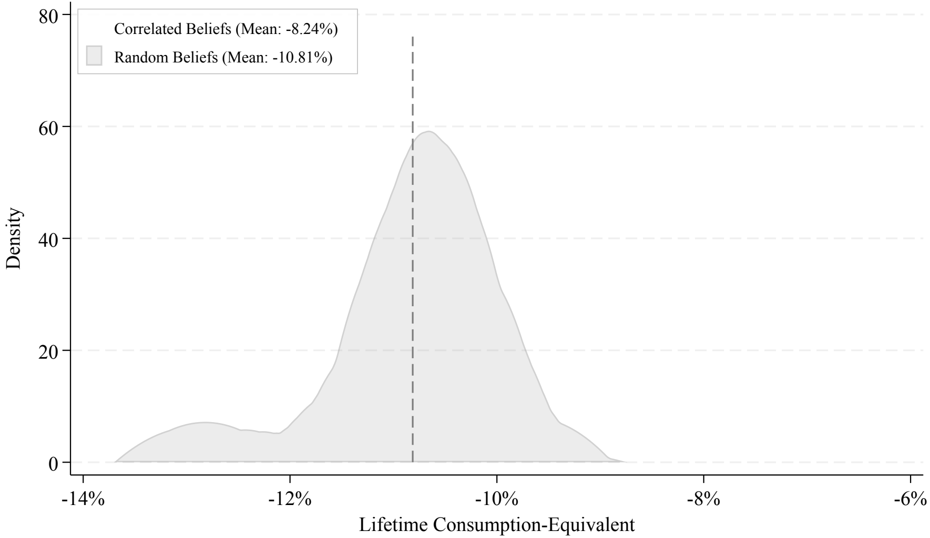
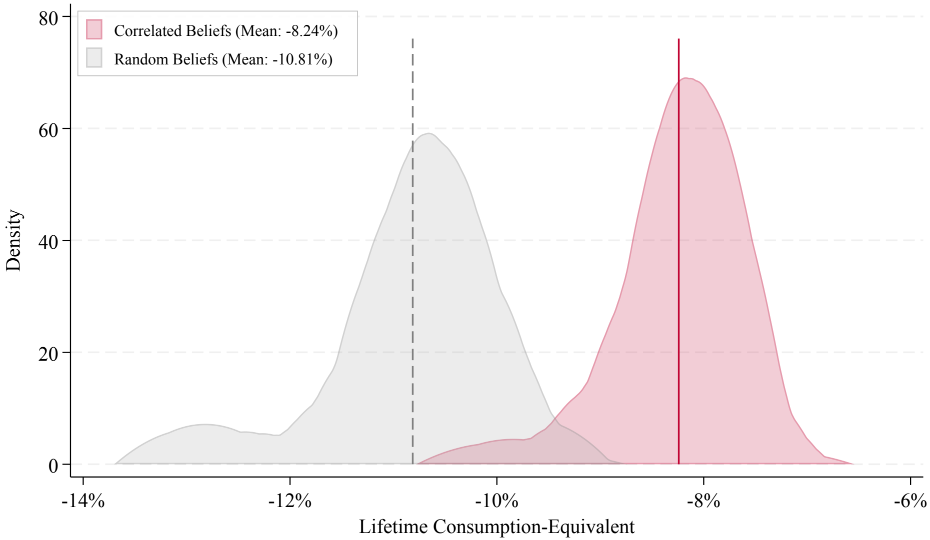
$\Rightarrow$ Difference = welfare effect of **belief correlations**.

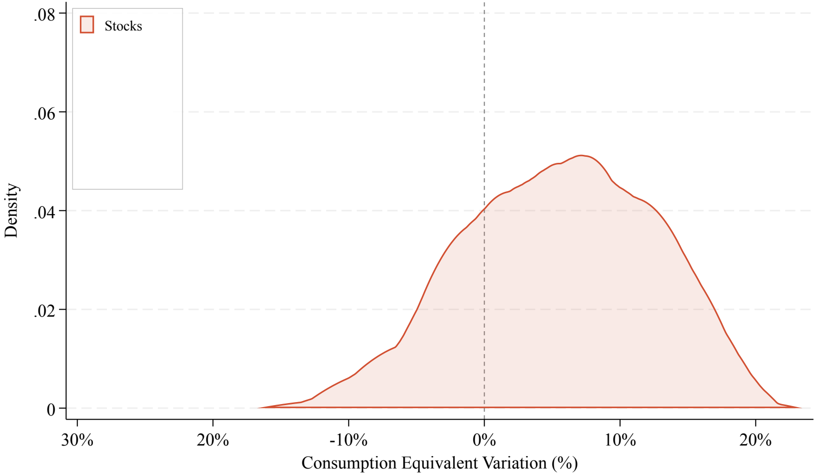
Finding 2. Correlations ↓ Welfare Cost of Belief Heterogeneity



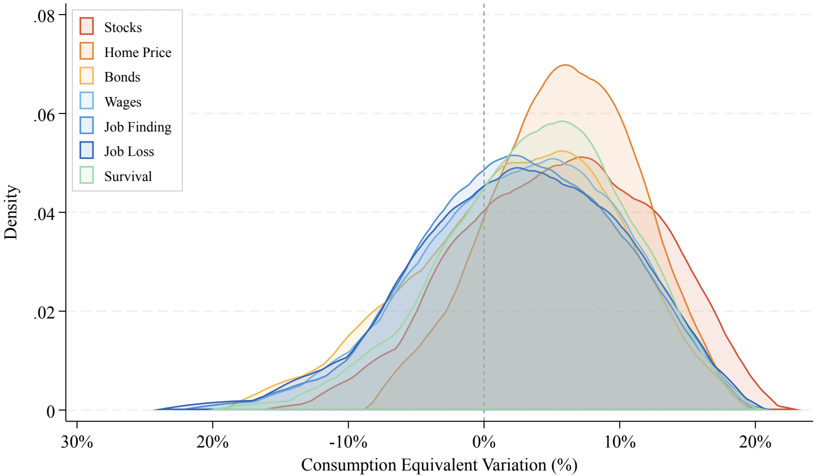
Finding 2. Correlations ↓ Welfare Cost of Belief Heterogeneity



Finding 3. Correcting Single Beliefs Can Backfire



Finding 3. Correcting Single Beliefs Can Backfire



- Roughly 1/4 of belief groups experience a welfare decline
- Outcomes worst for young, low income, low fin. lit. and pessimistic groups

Outline

Conclusion

Three main results.

- Belief correlations often **attenuate** the pass-through from beliefs to behavior
  $\rightarrow$ may help explain weak sensitivity of behavior to self-reported expectations
- Belief correlations **reduce** the welfare cost of belief heterogeneity by $\sim$ **25%**
- Correcting **single** beliefs can **reduce** welfare among pessimists

$\Rightarrow$ **Studying expectations one domain at a time misses this.**

Conclusion

Three main results.

- Belief correlations often **attenuate** the pass-through from beliefs to behavior
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- Belief correlations **reduce** the welfare cost of belief heterogeneity by $\sim$ **25%**
- Correcting **single** beliefs can **reduce** welfare among pessimists

$\Rightarrow$ **Studying expectations one domain at a time misses this.**

Open question. *Why* do belief correlations take *this* form?

Hypothesis. The most costly belief combinations may get corrected over time:

- Personal experience
- Social learning
- Transmission of adaptive heuristics

Static Portfolio Choice Decisions

- **Static mean-variance portfolio allocation**: two risky + one risk-free assets

$$\tilde{w}_1 - w_1^* = (\gamma \det(\Sigma))[\sigma_2^2(\mu_1 \epsilon_{\mathbf{1}} - r\epsilon_{\mathbf{r}}) - \sigma_{12}(\mu_2 \epsilon_{\mathbf{2}} - r\epsilon_{\mathbf{r}})]$$

- **Agents holds biased beliefs** about expected returns

Asset 1	Asset 2	Risk-free rate
$\widetilde{\mu_1} = (1+\epsilon_{\mathbf{1}})\,\mu_1$	$\widetilde{\mu_2} = (1+\epsilon_{\mathbf{2}})\,\mu_2$	$\widetilde{r} = (1+\epsilon_{\mathbf{r}})\,r$

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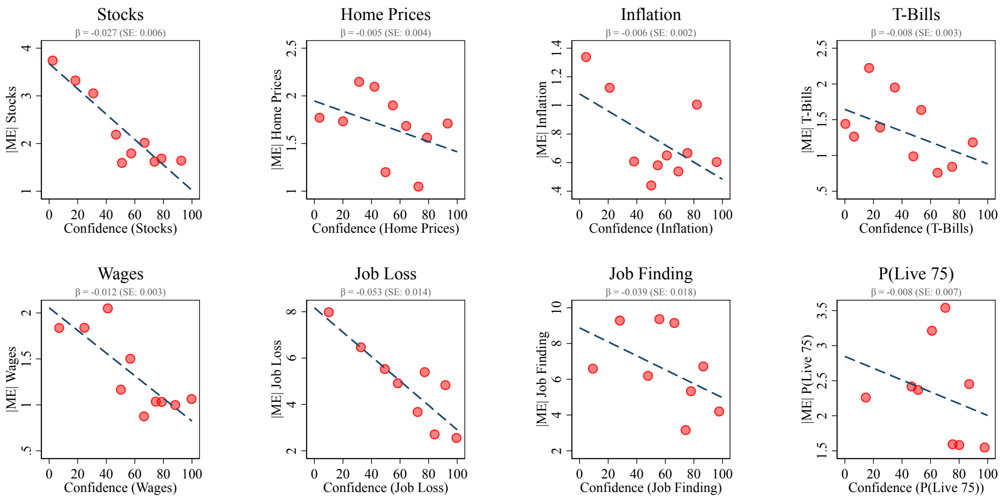
Measurement Error and Confidence



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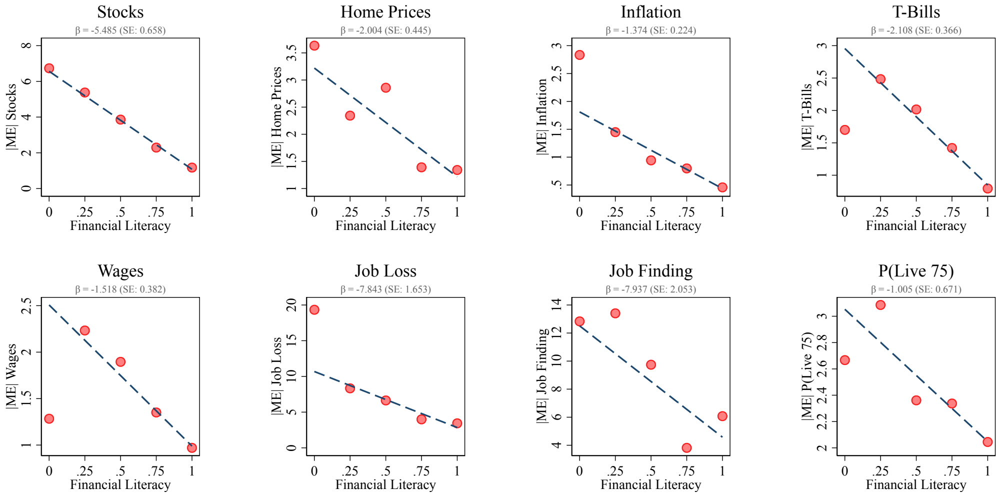
Measurement Error and Financial Literacy



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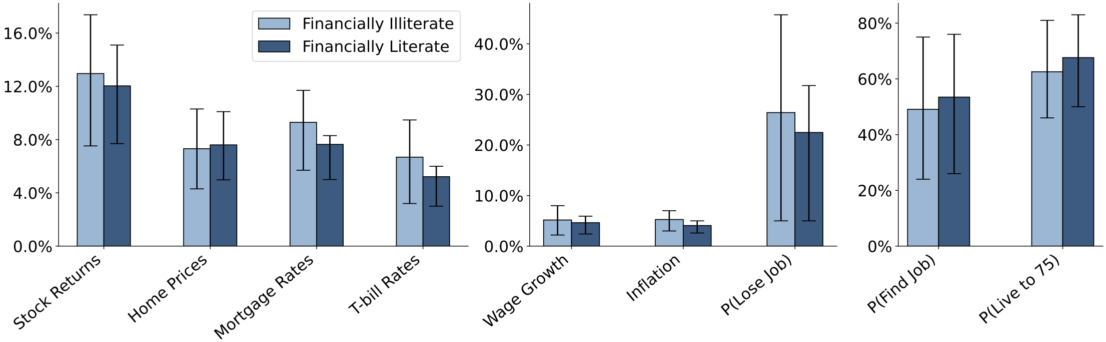
Expectations by Financial Literacy



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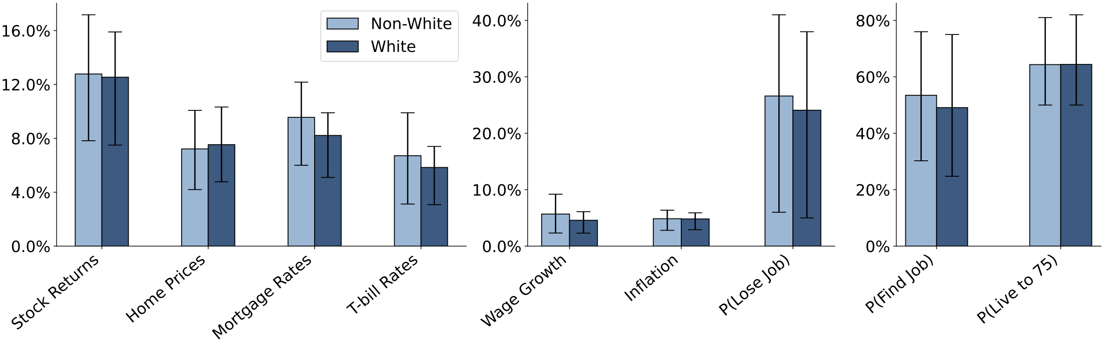
Expectations by Race



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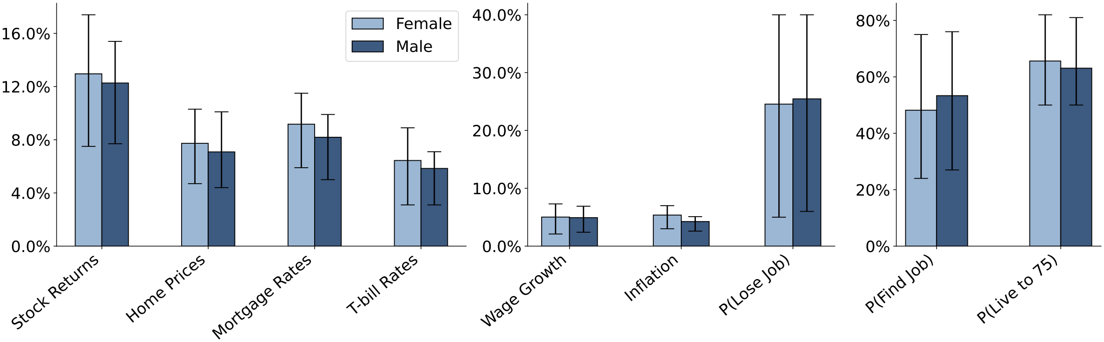
Expectations by Gender



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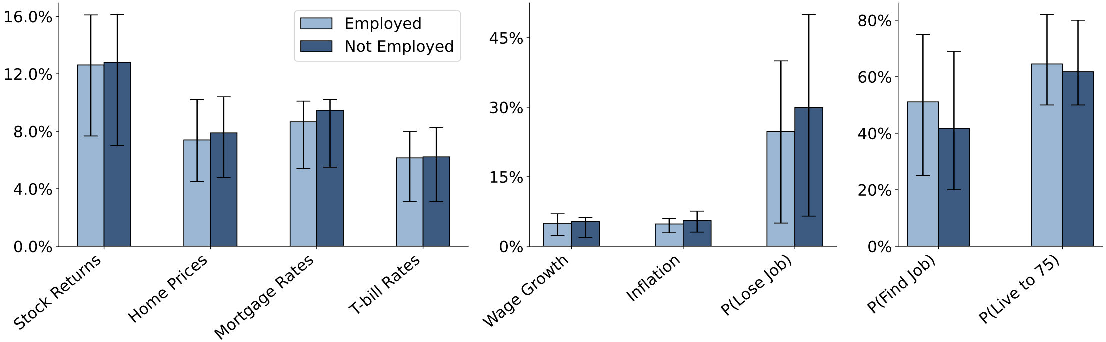
Expectations by Employment Status



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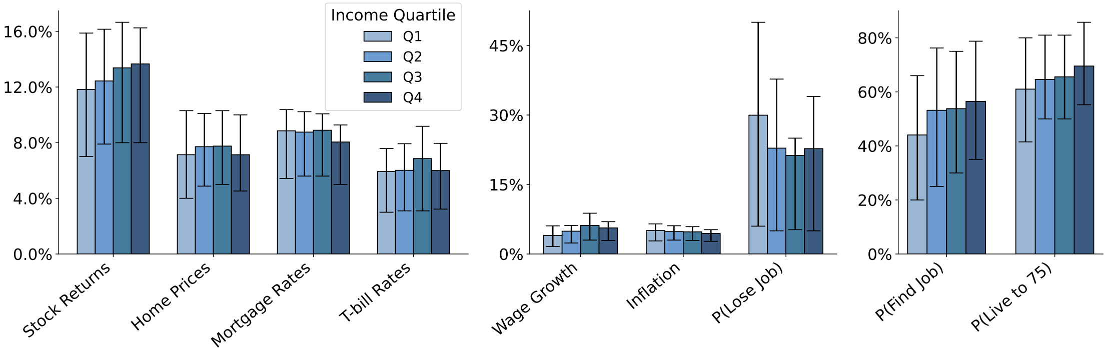
Expectations by Income



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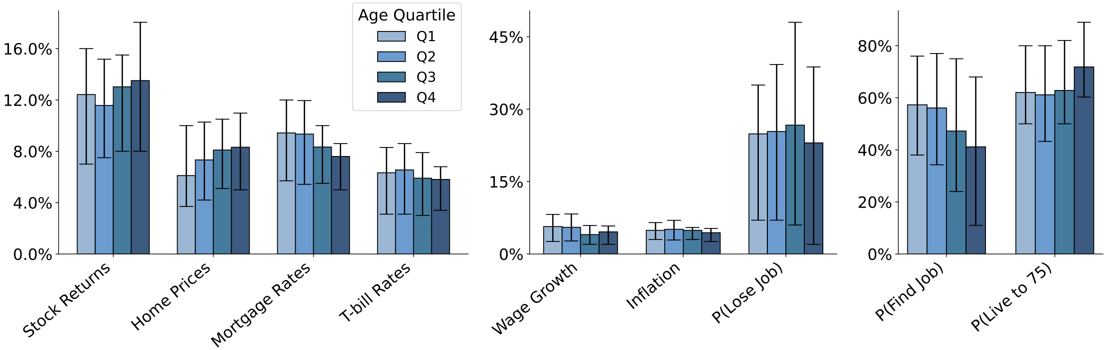
Expectations by Age



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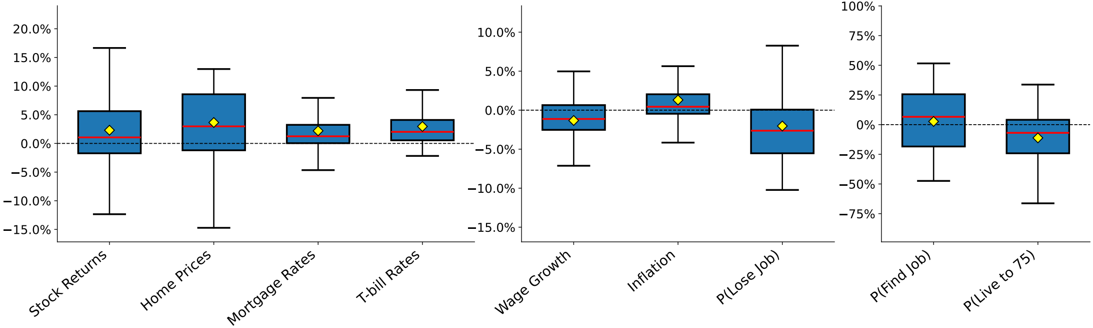
Average Bias



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Survey: Memory Mortgage Rates

Over the **past 10 years**, what do you think the **average 30-year fixed mortgage interest rate** was in the U.S.?

0% 5% 10% 15% 20%

10 year avg.

Back

Survey: Typical Life Expectancy

Among individuals of your same age and sex, what do you think is the **percent chance (from 0% to 100%) of living to age 75 or older**?

0% 20% 40% 60% 80% 100%

Probability

Back

Survey: Memory Inflation

Over the **past 10 years**, what do you think **the average annual inflation rate** was? (the percent change in the prices of goods and services over a year)

-10%	-5%	0%	5%	10%	15%

10 year avg.

Back

Survey: Typical Job Finding

In the U.S. over the **past 10 years**, what do you think the **average percentage of unemployed individuals who found a new job** within one year of being laid off or fired was?

0% 25% 50% 75% 100%

10 year avg.

Back

Survey: Typical Job Loss

In the U.S. over the **past 10 years**, what do you think the **average probability that an employed individual was laid off or fired** was in a typical year?

0% 5% 10% 15% 20%

10 year avg.

Back

Survey: Typical Wages

Over the **past 10 years**, what do you think has been the **average annual percentage change in the wage** of a typical worker in the U.S.?

-10% -5% 0% 5% 10%

10 year avg.

Back

Survey: Memory T-bill

Over the **past 10 years**, what do you think the **average annual return on a 1-year Treasury bill** was in the U.S.?
(a 1-year Treasury bill is a short-term U.S. government bond with virtually no default risk)

0% 5% 10% 15% 20%

10 year avg.

Back

Survey: Memory Home Prices

Over the **past 10 years**, what do you think the **average annual percentage change in U.S. home prices** was?"

-20% -10% 0% 10% 20%

10 year avg. annual change

Back

Survey: Memory Stocks

Over the **past 10 years**, what do you think the **average annual return was for an investment in the S&P 500** stock market index?
(the S&P 500 tracks the performance of 500 of the largest publicly traded companies in the U.S.)

-50% -25% 0% 25% 50%

10 year avg. annual return

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Survey: Psychological Measures

On a scale from 1 (Strongly Disagree) to 5 (Strongly Agree), how much do you agree with the following statements?

Strongly Disagree		Neutral		Strongly Agree
1	2	3	4	5

In uncertain times, I usually expect the best.

I always look on the bright side of things.

I'm always optimistic about my future.

→

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Survey: Attention Checks

Attention Check

You will flip a fair coin 10 times. You earn **$1 for each time it lands on tails**.

- The **minimum** you can earn is **$0** (if all flips are heads).
- The **maximum** is **$10** (if all flips are tails).
- The **most likely** amount is **$5** (since there's a 50% chance of tails).

Use the sliders below to indicate the:

- Minimum possible earnings
- Maximum possible earnings
- Most likely earnings

$0 $2 $4 $6 $8 $10

Minimum possible earnings

Maximum possible earnings

Most likely earnings

Back

Survey: Job Finding Expectations

Thinking about the **remainder of your career**, imagine you lose your job at some point. What do you think is the **percent chance (from 0% to 100%) that you would find a new job** with about the same pay within one year of losing your job?

0% 25% 50% 75% 100%

Probability

Back

Survey: Job Loss Expectations

Thinking about the **remainder of your career**, what do you expect is the **percent chance (from 0% to 100%) you lose your job** in a given year due to being laid off or fired?

0% 25% 50% 75% 100%

Probability

Back

Survey: Wage Growth Expectations

For the **remainder of your career** what do you expect your **average annual percent change in wages** will be? (Please consider your total wage changes from year to year, whether increases from raises or promotions, or decreases from cuts.)

-25%	-15%	-5%	5%	15%	25%

Avg. percent change

Back

Survey: Life Expectancy

What do you believe is the **percent chance (from 0% to 100%) you live to age 75** or older?

0% 20% 40% 60% 80% 100%

Probability

Back

Survey: Inflation Expectations

Over the next **20 years**, what do you believe the **average U.S annual inflation rate** will be? (i.e., the percent change in the prices of goods and services over a year)?

-10% -5% 0% 5% 10% 15%

Avg. annual rate

Back

Survey: T-bill Expectations

Over the next **20 years**, what do you believe the **average annual return on a 1-year Treasury bill** will be in the U.S.? (a 1-year Treasury bill is a short-term U.S. government bond with virtually no default risk)

0% 5% 10% 15% 20% 25%

Avg. annual return

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Survey: Mortgage Rate Expectations

Over the next **20 years**, what do you believe the **average 30-year fixed mortgage interest rate** will be in the U.S.?

0%	5%	10%	15%	20%	25%

Avg. Rate

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Survey: Housing Expectations

Imagine as a long term investment you invest in a **U.S. housing market index fund** (a fund that simply follows the entire U.S. housing market). If you held that fund for the next **20 years**, what **average annual return** would you expect?

-20% -10% 0% 10% 20%

Avg. annual return

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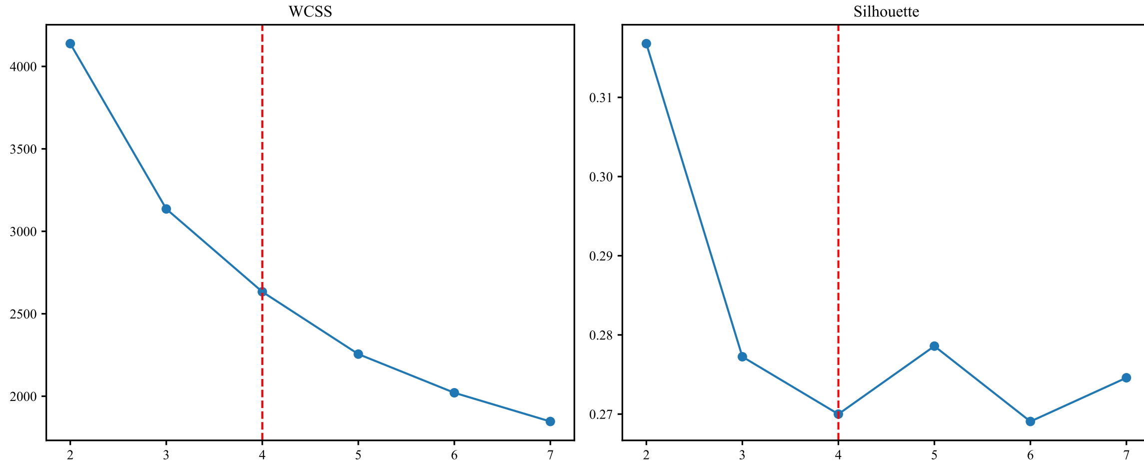
Cluster Scores



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